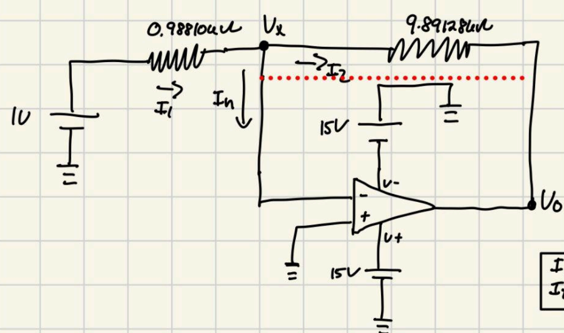
Frank Tamburro

ECE 121 Lab #4

Experiment 1

Experiment 1		Calculated	Measured	Simulated		Value	Actual Value
Vs=1V	Vo	Vo=-10.01V	Vo=0.010mV	Vo=14.815V		Value	Actual Value
	Vx	Vx=0V	Vx=0.003mV	Vx=-7.7462	Resistor 1	1k	0.98810k
	I1	I1=1.012mA	I1=1.0246mA	I1=-2.2809mA	Resistor 2	10k	9.89128K
	I2	I2=1.012mA	I2=1.0237mA	I2=-2.2809mA			
Vs=5V	Vo	Vo=-50.0519V	Vo=0.019mV	Vo=14.815V			
	Vx	Vx=0V	Vx=0.007mV	Vx=-7.7462			
	I1	I1=5.0602mA	I1=1.6472mA	I1=-2.2809mA			
	I2	I2=5.0602mA	I2=1.6471mA	I2=-2.2809mA			



$V_p = V_n = 0V$

$I_n = I_p = 0 A$

$V_n = V_x = 0V$

$I_1 - I_2 - I_n = 0$

$I_1 - I_2 - 0 = 0$

$I_1 = I_2$

1V

$\boxed{I_1 = 1.012mA \quad I_2 = 1.012mA}$

$\frac{1-0}{0.98810k\Omega} = \frac{0-V_o}{9.89128k}$

$1.012mA = \frac{-V_o}{9.89128k} \cdot 9.89128k$

$10.01V = -V_o$

$V_o = -10.01V$

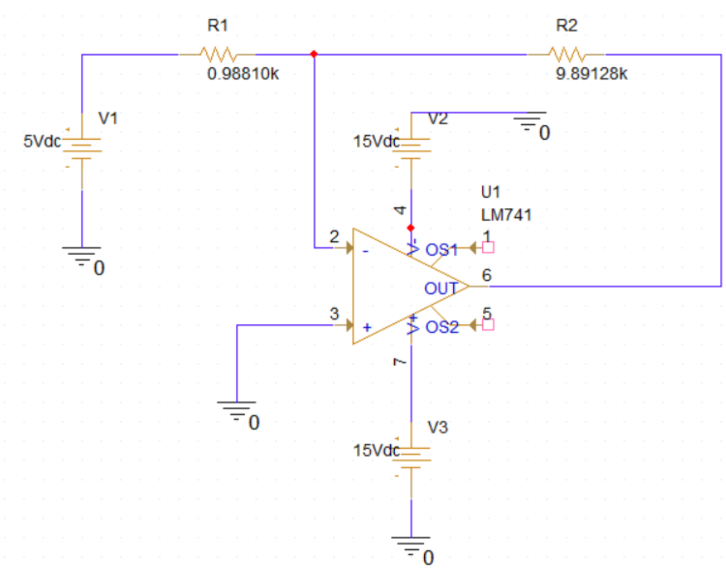
5V

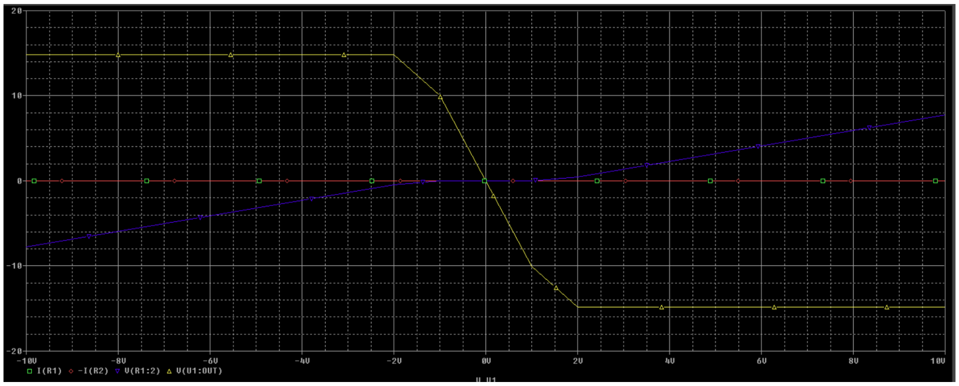
$\boxed{I_1 = 5.0602mA \quad I_2 = 5.0602mA}$

$\frac{5-0}{0.98810k\Omega} = \frac{0-V_o}{9.89128k\Omega}$

$5.0602mA = \frac{-V_o}{9.89128k}$

$V_o = -50.0519V$

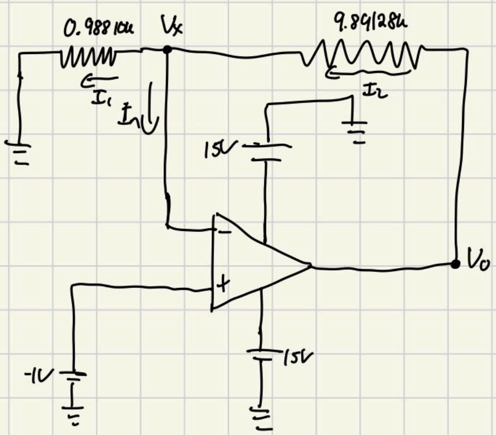




	Trace Color	Trace Name	Y1	Y2	Y1 - Y2		Y1(Cursor1) - Y2(Cursor2)		0.000		
		X Values	-10.000	-10.000	0.000		Y1 - Y1(Cursor1)	Y2 - Y2(Cursor2)	Max Y	Min Y	Avg Y
	CURSOR 1,2	V(U1:OUT)	14.815	14.815	0.000		0.000	0.000	14.815	14.815	14.815
		V(R1:2)	-7.7462	-7.7462	0.000		-22.561	-22.561	-7.7462	-7.7462	-7.7462
		I(R1)	-2.2809m	-2.2809m	0.000		-14.817	-14.817	-2.2809m	-2.2809m	-2.2809m
		I(R2)	-2.2809m	-2.2809m	0.000		-14.817	-14.817	-2.2809m	-2.2809m	-2.2809m

Experiment 2

Experiment 2		Calculated	Measured	Simulated		Value	Actual Value
Vs=-1V	Vo	Vo=8.9101V	Vo=-0.008mV	Vo=-14.815V			
	Vx	Vx=-1V	Vx=-0.005mV	Vx=-1.3457V	Resistor 1	1K	0.99810K
	I1	I1=0.50602mA	I1=-1.0009mA	I1=-1.3619mA	Resistor 2	10K	9.89128K
	I2	I2=0.50602mA	I2=-1.0143mA	I2=1.3618mA			
Vs=0.5V	Vo	Vo=4.505V	Vo=0.002mV	Vo=-14.815V			
	Vx	Vx=0.5V	Vx=0.014mV	Vx=-1.3457V			
	I1	I1=0.50602mA	I1=-0.5102mA	I1=-1.3619mA			
	I2	I2=0.50602mA	I2=-0.5119mA	I2=1.3618mA			



-1V

$V_p = V_n = -1V$

$V_x = V_n = -1V$

$I_2 - I_1 - I_n = 0$

$I_2 - I_1 - 0 = 0$

$I_2 = I_1$

$\frac{V_o + 1V}{9.89128k} = \frac{-1-0}{0.98810k}$

$\frac{V_o + 1V}{9.89128k} = \frac{-1}{0.98810}$

$\frac{V_o + 1V}{9.89128k} = 1.0019mA \cdot 9.89128k$

$V_o + 1V = 9.9101V - 1V$

$V_o = 8.9101V$

-1V

$\boxed{I_1 = 1.0019mA \quad I_2 = 1.0019mA}$

0.5V

$V_p = V_n = 0.5V$

$V_x = V_n = 0.5V$

$I_2 - I_1 - I_n = 0$

$I_2 - I_1 - 0 = 0$

$I_2 = I_1$

$\frac{V_o - 0.5}{9.89128k} = \frac{0.5-0}{0.98810k}$

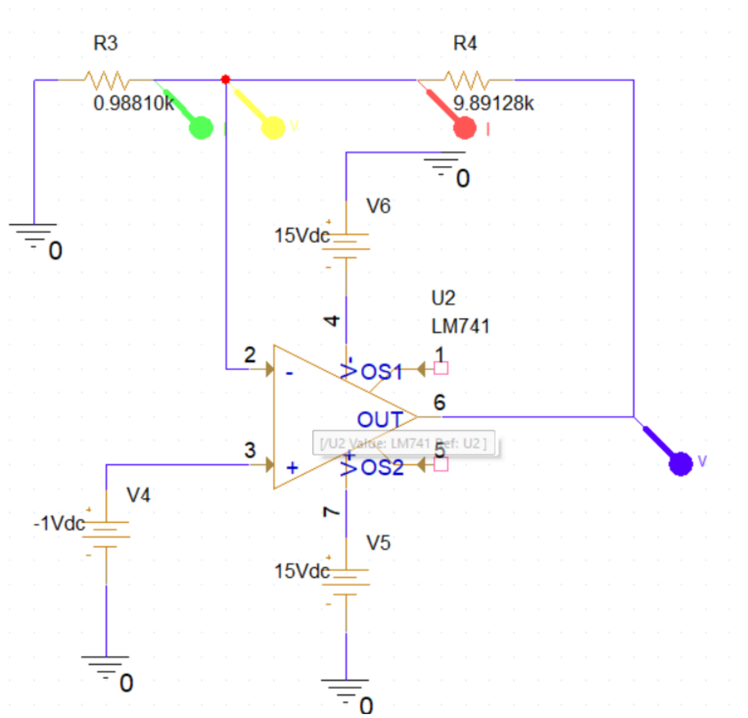
$9.89128k\left(\frac{V_o - 0.5}{9.89128k} = 0.50602mA\right)$

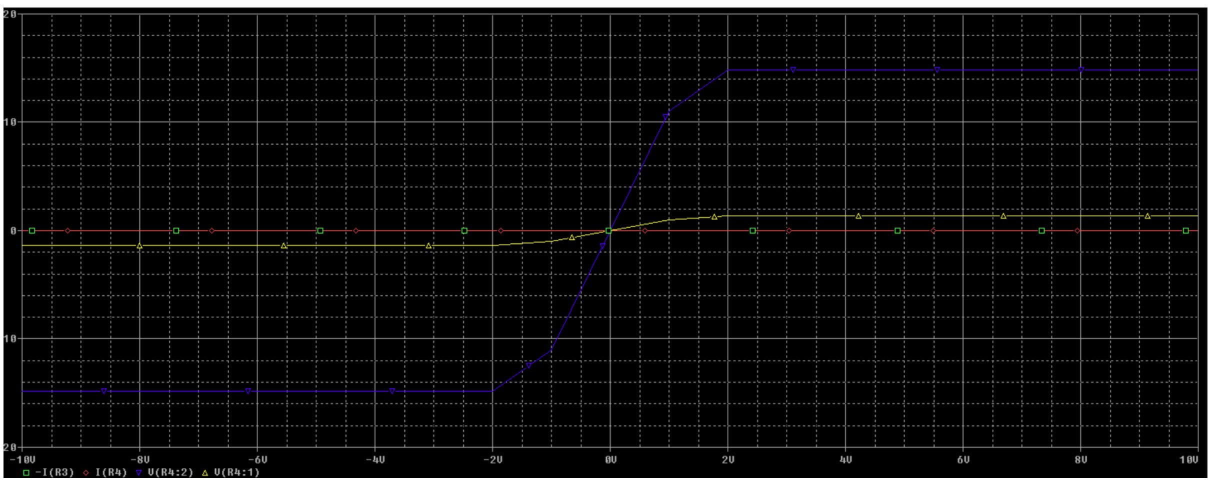
$V_o - 0.5 = 5.005V$

$V_o = 4.505V$

0.5V

$\boxed{I_1 = 0.50602mA \quad I_2 = 0.50602mA}$





		X Values	-10.000	-10.000	0.000		Y1 - Y1(Cursor1)	Y2 - Y2(Cursor2)	Max Y	Min Y	Avg Y
	CURSOR 1,2	-I(R3)	-1.3619m	-1.3619m	0.000		0.000	0.000	-1.3619m	-1.3619m	-1.3619m
		I(R4)	1.3618m	1.3618m	0.000		2.7237m	2.7237m	1.3618m	1.3618m	1.3618m
		V(R4:2)	-14.815	-14.815	0.000		-14.814	-14.814	-14.815	-14.815	-14.815
		V(R4:1)	-1.3457	-1.3457	0.000		-1.3444	-1.3444	-1.3457	-1.3457	-1.3457